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OPERATIONS RESEARCH 415 PART IV: QUEUING THEORY

LECTURE 29: THREE WELL-KNOWN QUEUING SYSTEMS

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Recall, from Lecture 26, that

$$P_{i,j}(t + \Delta t) = \Delta t [\lambda_{j-1} P_{i,j-1}(t) + \mu_{j+1} P_{i,j+1}(t) - \mu_j P_{i,j}(t) - \lambda_j P_{i,j}(t)] + P_{i,j}(t) + o(\Delta t)$$

Letting

$$\lambda_j = \lambda, \quad j = 0, 1, 2, \dots$$

$$\mu_j = j\mu, \quad j = 0, 1, \dots, s$$

$$\mu_j = s\mu, \quad j = s + 1, s + 2, \dots$$

this leads to a system of differential equations. Letting $P_{i,j}(t) = \pi_j$ in the steady state as $t \rightarrow \infty$ and noting that $\pi'_j = 0$, this system reduces to a system of algebraic equations which can be solved to find

$$\pi_j = \begin{cases} 1 / \left(\sum_{i=0}^{s-1} \frac{(s\rho)^i}{i!} + \frac{(s\rho)^s}{s!(1-\rho)} \right), & j = 0 \\ \frac{(s\rho)^j \pi_0}{j!}, & j = 1, 2, \dots, s \\ \frac{(s\rho)^j \pi_0}{s! s^{j-s}}, & j = s + 1, s + 2, \dots \end{cases}$$

where $\rho = \frac{\lambda}{s\mu} < 1$ (otherwise no steady state exists).

The System $M/M/s/GD/\infty/\infty$ (pp. 1087–1088)

The steady state probability that all servers are busy is

$$P(j \geq s) = \frac{(s\rho)^s \pi_0}{s!(1 - \rho)}$$

$P(j \geq s)$ for the $M/M/s/GD/\infty/\infty$ Queuing System

ρ	$s = 2$	$s = 3$	$s = 4$	$s = 5$	$s = 6$	$s = 7$
.10	.02	.00	.00	.00	.00	.00
.20	.07	.02	.00	.00	.00	.00
.30	.14	.07	.04	.02	.01	.00
.40	.23	.14	.09	.06	.04	.03
.50	.33	.24	.17	.13	.10	.08
.55	.39	.29	.23	.18	.14	.11
.60	.45	.35	.29	.24	.20	.17
.65	.51	.42	.35	.30	.26	.21
.70	.57	.51	.43	.38	.34	.30
.75	.64	.57	.51	.46	.42	.39
.80	.71	.65	.60	.55	.52	.49
.85	.78	.73	.69	.65	.62	.60
.90	.85	.83	.79	.76	.74	.72
.95	.92	.91	.89	.88	.87	.85

Furthermore,

$$L_q = \frac{P(j \geq s)\rho}{1 - \rho}$$

$$W_q = \frac{L_q}{\lambda}$$

$$L_s = \frac{\lambda}{\mu}$$

$$W_s = \frac{1}{\mu}$$

$$L = L_q + \frac{\lambda}{\mu}$$

$$W = \frac{P(j \geq s)}{s\mu - \lambda} + \frac{1}{\mu}$$

Example 9 (p. 1089)

Consider a bank with two tellers.

An average of 80 customers per hour arrive at the bank and wait in a single line for an idle teller.

The average time it takes to serve a customer is 1.2 minutes.

Assume that inter-arrival times and service times are exponentially distributed.

- 1 Determine the expected number of customers present in the bank.
- 2 Determine the expected length of time a customer spends in the bank.
- 3 Determine the fraction of time that a particular teller is idle.

There are many examples of systems, such as $M/G/\infty/GD/\infty/\infty$ or $GI/G/\infty/GD/\infty/\infty$, in which a customer never has to wait for service to begin.

In such a system, the customer's entire stay in the system may be thought of as his or her service time.

Since a customer never has to wait for service, there is, in essence, a server available for each arrival, and we may think of such a system as an infinite-server (or self-service) system.

Examples of Infinite-Server Queuing Systems

Situation	Arrival	Service Time (time in system)	State of System
Industry	Firm enters industry	Time until firm leaves industry	Number of firms in industry
College program	Student enters program	Time student remains in program	Number of students in program

In such a system $L = \frac{\lambda}{\mu}$ and $W = \frac{1}{\mu}$.

The above relations hold, even without assumptions of exponentiality with respect to inter-arrival and/or service times, *i.e.* for the system $GI/G/\infty/GD/\infty/\infty$.

However, if the inter-arrival times are exponentially distributed (*i.e.* for the system $M/G/\infty/GD/\infty/\infty$), then the steady state probability that there are j customers in the system is Poisson distributed with parameter $\frac{\lambda}{\mu}$, that is

$$\pi_j = \frac{\left(\frac{\lambda}{\mu}\right)^j e^{-\lambda/\mu}}{j!}, \quad j = 0, 1, 2, \dots$$

Example 11 (p. 1096)

During a given year, an average of three ice cream shops open up in Smalltown.

The average time that an ice cream shop remains in business is ten years.

- 1 On January 1, 2525, what is the average number of ice cream shops that you would find in Smalltown?
- 2 If the times between opening of ice cream shops is exponentially distributed, what is the probability that on January 1, 2525, there will be 25 ice cream shops in Smalltown?